

Wasserstein Gradient Flow Structure in Mean-Field Oscillator Ising Machines

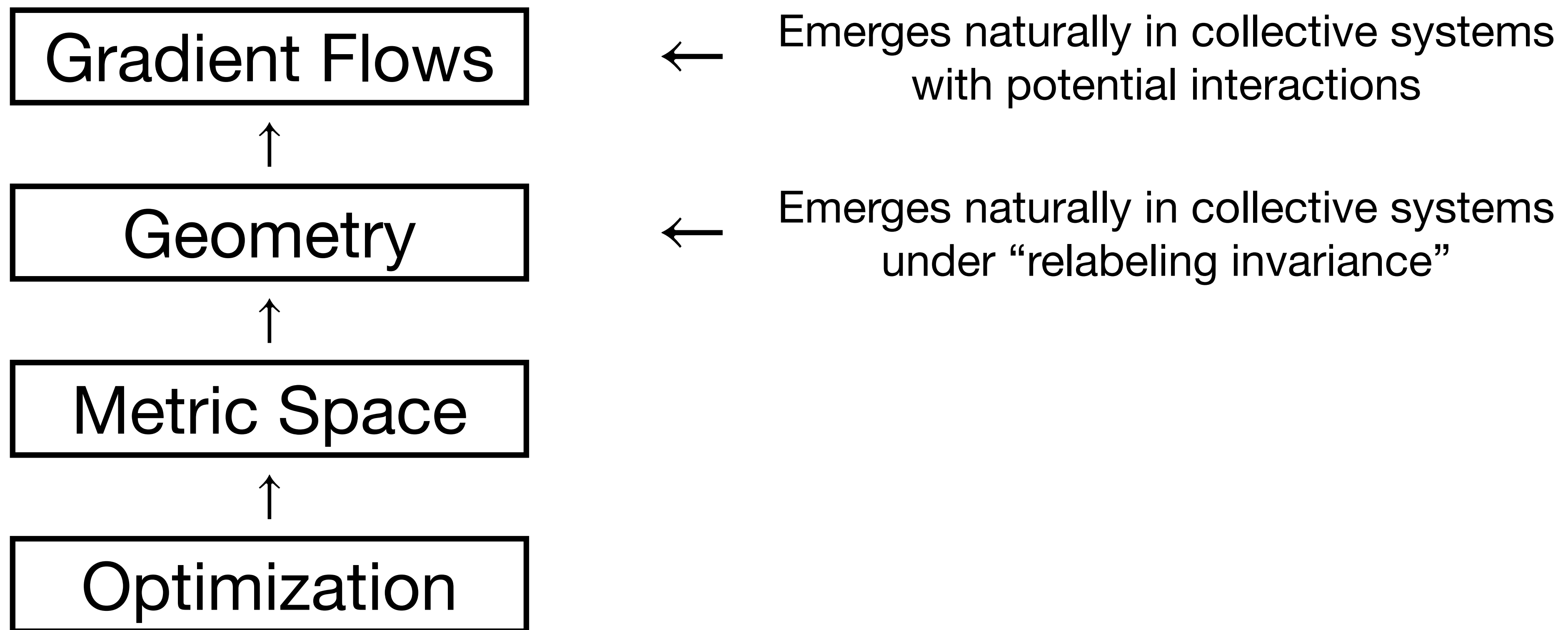
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Context: A Case Study in OT/Gradient Flows

Four Levels of Optimal Transport:



Background: Gradient Flows

- Classical (Euclidean) gradient flow: $\dot{x} = -\nabla E(x)$
- Can be extended to more general settings (e.g. Riemannian manifolds)
- Gradients must be defined w/r/t particular geometry:

$$\langle \nabla E(x), v \rangle_x = DE(x)[v] = \lim_{h \rightarrow 0} \frac{E(x + hv) - E(x)}{h}$$

- Energy + Geometry $\rightarrow$ Dynamics (Gradient Flow)
- Also exist notions of gradient flows in metric spaces [1]

Background: Gradient Flows

- Gradient flows are highly structured objects
- For sufficiently nice E , we expect:
 - Dynamics are well-posed
 - All trajectories converge to fixed points ($\nabla E(x) = 0$)
 - Most trajectories converge to local minima ($\nabla^2 E(x) > 0$)
 - ∇E and $\nabla^2 E$ admit closed-form expressions via calculus of variations
 - Allows for a complete classification of limit behavior

Background: Gradient Flows

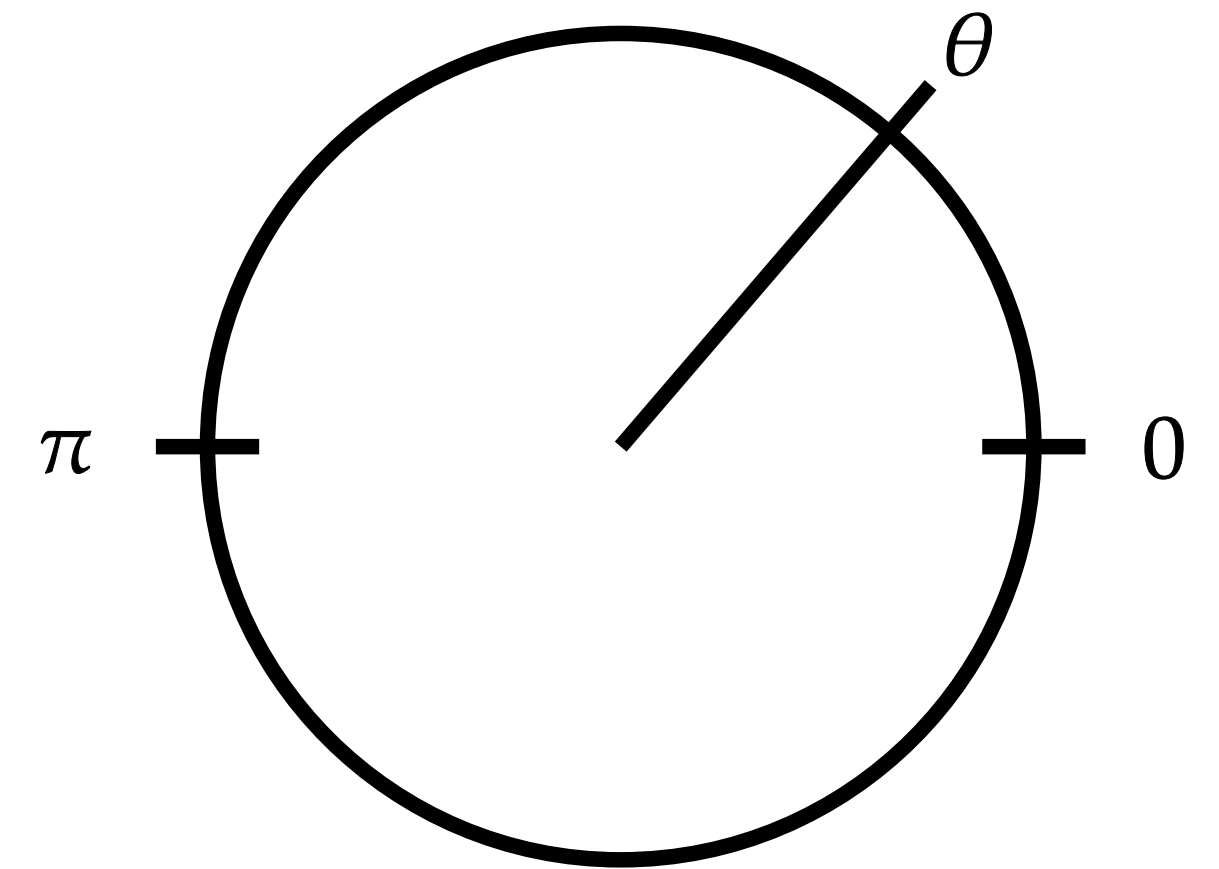
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*The devil is always in the details

Oscillator Ising Machines (OIMs)

Discrete Problem:

$$\min_{\sigma \in \{\pm 1\}^n} \frac{1}{2} \sum_{i,j} A_{ij} \sigma_i \sigma_j$$



Continuous Problem:

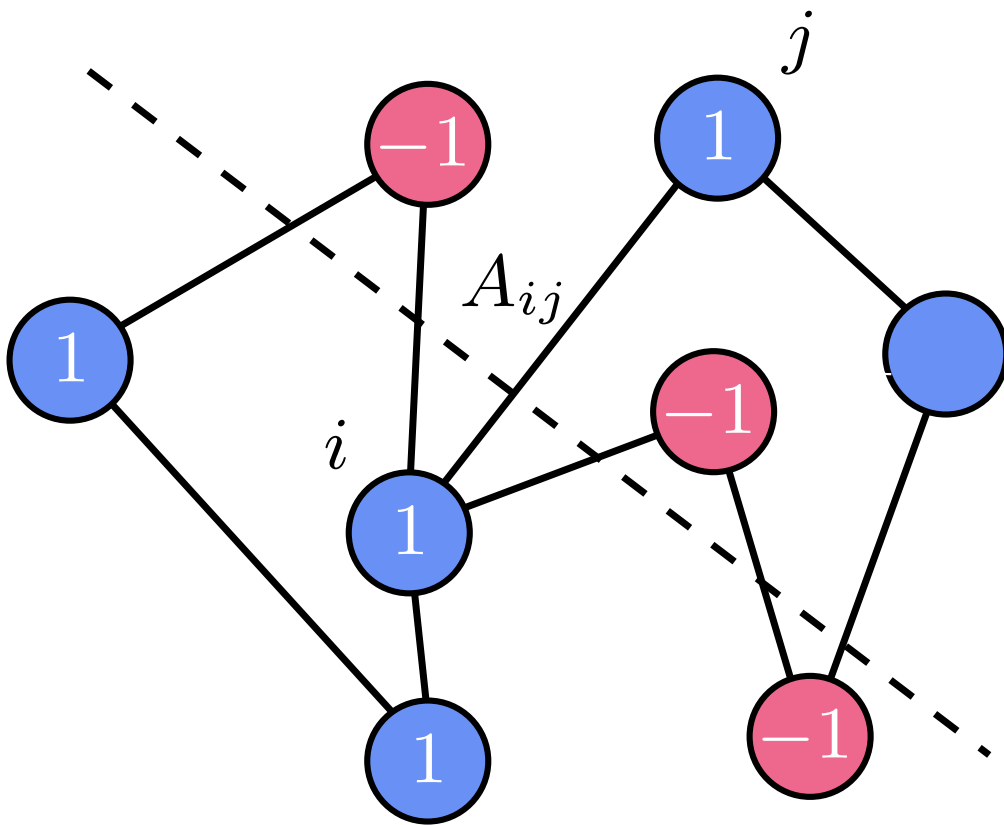
$$\min_{\theta \in \mathbb{T}^n} \frac{1}{2} \sum_{i,j} A_{ij} \cos(\theta_i - \theta_j) - \frac{K_s}{2} \sum_i \cos(2\theta_i)$$

Gradient Descent:

$$\dot{\theta}_i = \sum_j A_{ij} \sin(\theta_i - \theta_j) - K_s \sin(2\theta_i)$$

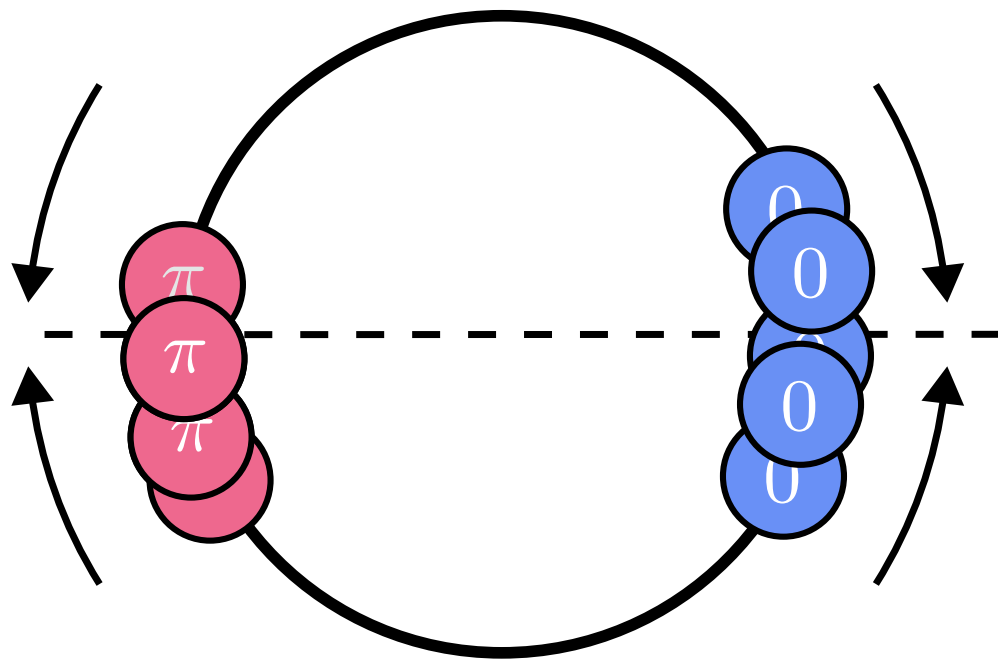
Oscillator Ising Machines (OIMs)

Max-Cut



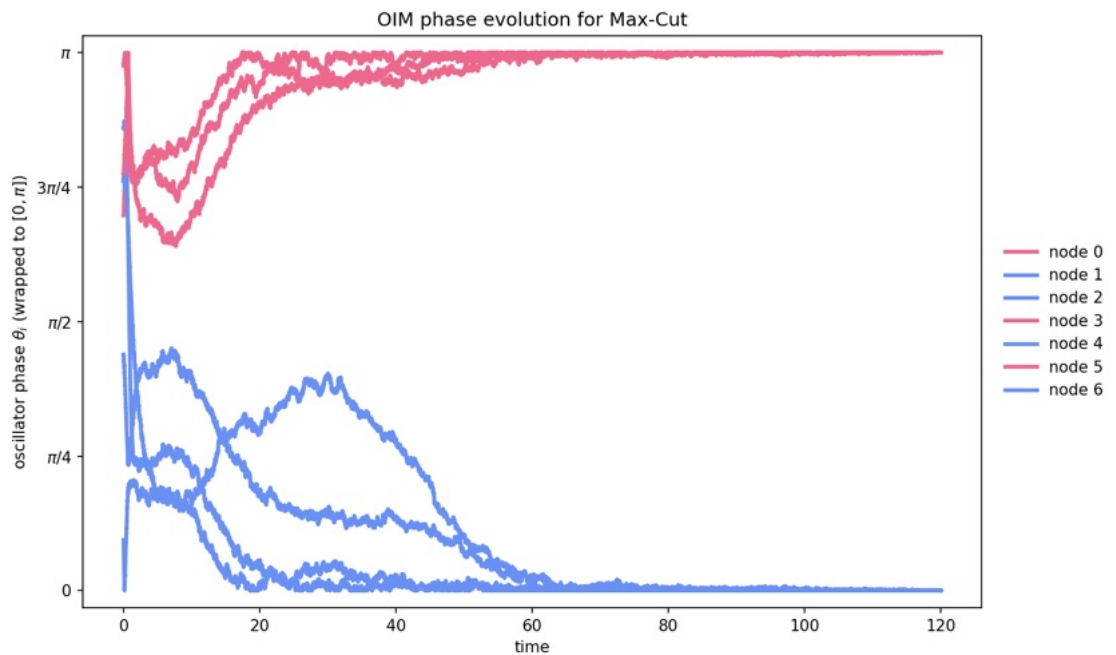
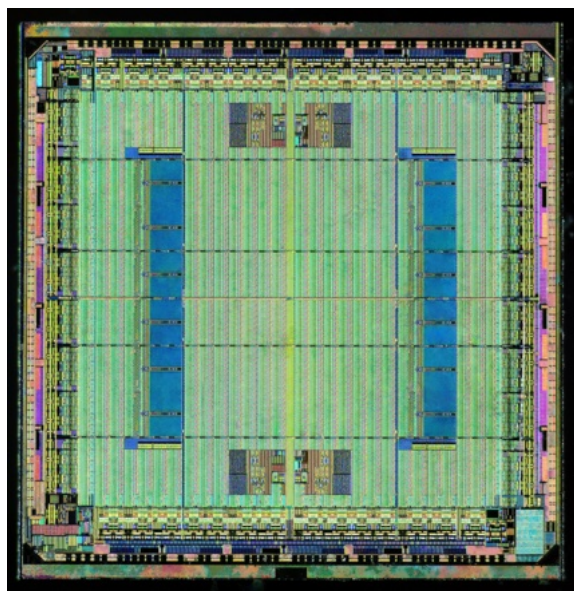
$$\min_{\sigma \in \Sigma^n} \frac{1}{2} \sum_{i,j} A_{ij} \sigma_i \sigma_j$$

Oscillator Ising Machines



$$\dot{\theta}_i = \sum_j A_{ij} \sin(\theta_i - \theta_j) - K_s \sin(2\theta_i)$$

Analog implementation using FPGA



Remainder of Talk

- Investigate mean field ($n \rightarrow \infty$) limits of these OIMs
- Two different kinds of mean field models:
 - Lagrangian (always valid)
 - Eulerian (valid under relabeling invariance)
- Both limits retain gradient flow structures
- Develop classification of fixed points in relabeling-invariant case
- Show numerical evidence that classification predicts behavior in certain large random networks

Lagrangian Mean-Field Limit

- Index set $\mathfrak{n} = \{1, \dots, n\}$ converges to $I = [0,1]$
- Adjacency matrix $A : \mathfrak{n} \times \mathfrak{n} \rightarrow \mathbb{R}$ converges to graphon $K : I \times I \rightarrow \mathbb{R}$
- Lagrangian energy and dynamics derived formally by passing to continuum:

$$E(\boldsymbol{\theta}) = \frac{1}{2} \iint K(x, y) \cos(\boldsymbol{\theta}(x) - \boldsymbol{\theta}(y)) \, dx \, dy - \frac{K_s}{2} \int \cos(2\boldsymbol{\theta}(x)) \, dx$$

$$\partial_t \boldsymbol{\theta}(x) = \int K(x, y) \sin(\boldsymbol{\theta}(x) - \boldsymbol{\theta}(y)) \, dy - K_s \sin(2\boldsymbol{\theta}(x))$$

- State $\boldsymbol{\theta} : I \rightarrow \mathbb{T}$ called the “phase function”
- Gradient flow w/r/t L^2 Riemannian structure on space of phase functions Θ

Eulerian-Lagrangian Correspondence

- Form phase density $\rho \in \mathcal{P}(\mathbb{T})$ by pushforward:

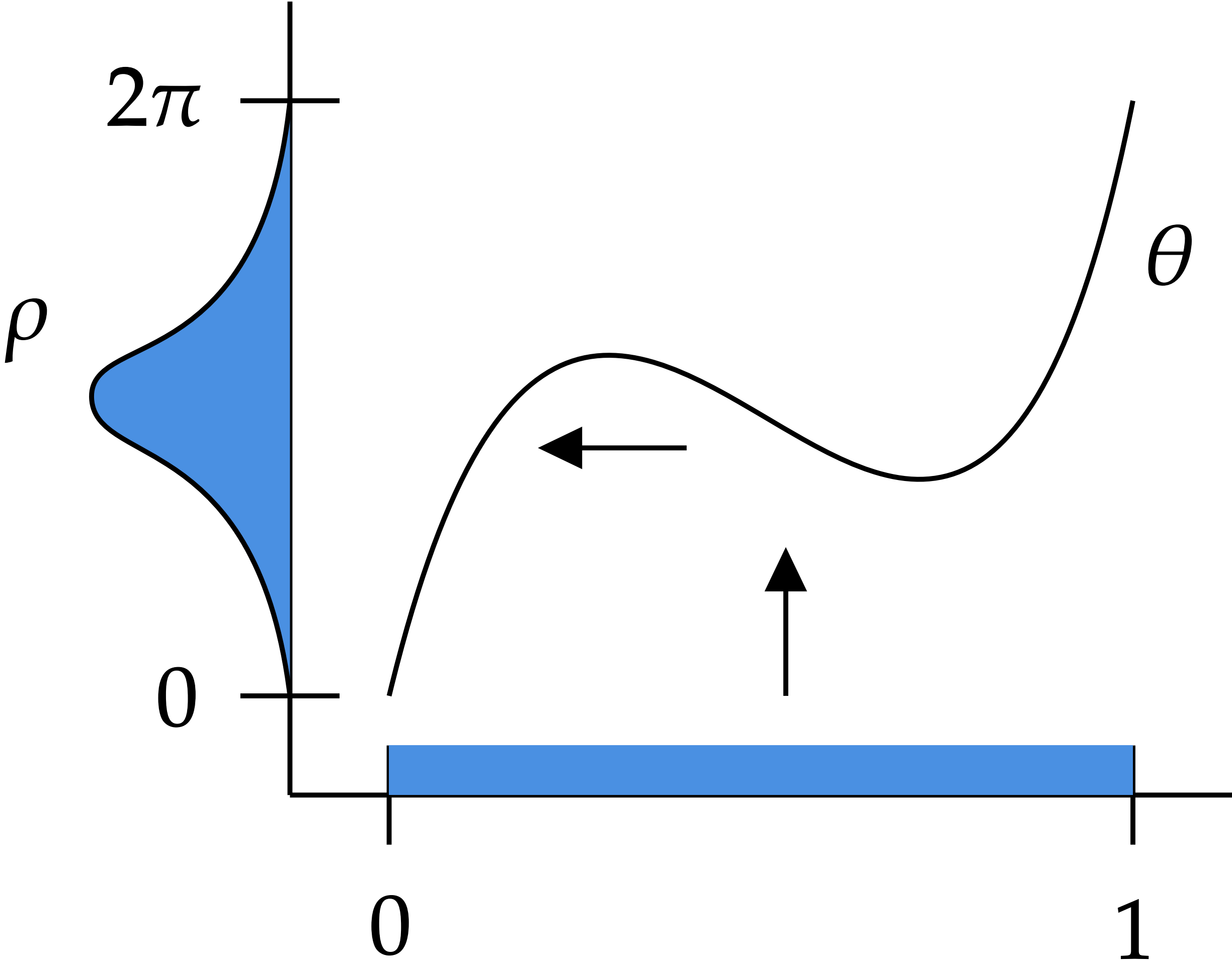
$$\rho = \theta_{\#} \mathbf{1}_I \quad \Leftrightarrow \quad \int_A \rho = \int_{\theta^{-1}(A)} 1$$

- Defines map $M : \Theta \rightarrow \mathcal{P}(\mathbb{T})$
- M is many-to-one:

$$\theta_{\#} \mathbf{1}_I = \phi_{\#} \mathbf{1}_I \quad \Leftrightarrow \quad \exists \omega : I \rightarrow I \text{ s.t. } \theta = \phi \circ \omega, \quad \omega_{\#} \mathbf{1}_I = \mathbf{1}_I$$

- Identifies equivalence classes of “relabeling equivalent” phase functions
- Space of phase densities identified with quotient: $\mathcal{P}(\mathbb{T}) = \Theta / \sim$

Eulerian-Lagrangian Correspondence



Eulerian Mean-Field Limit

- Phase density ρ always defined but does not always form a state
- ρ forms a state $\Leftrightarrow$ dynamics $\partial_t \theta$ can be pushed forward to $\partial_t \rho$
 - $\Leftrightarrow$ dynamics are invariant on relabeling equivalence classes
 - $\Leftrightarrow$ dynamics factor over $\mathbb{T}$: $\partial_t \theta(x) = v(\theta(x))$
 - $\Leftrightarrow K(x, y) \equiv K = \text{constant}$

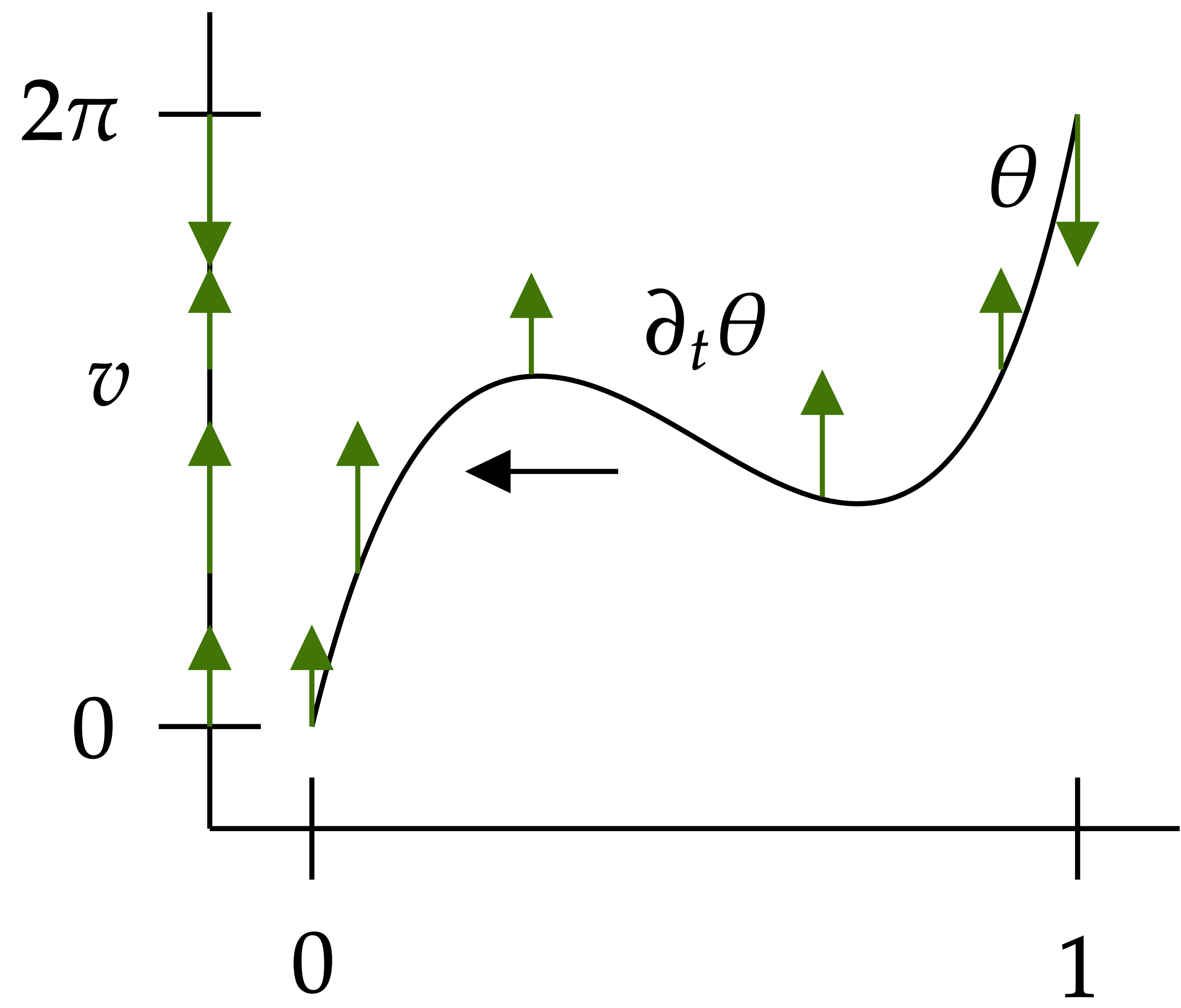
- In this case, Eulerian energy and dynamics given by:

$$E(\rho) = \frac{K}{2} \iint \cos(\theta - \phi) \rho(\theta) \rho(\phi) d\theta d\phi - \frac{K_s}{2} \int \cos(2\theta) \rho(\theta) d\theta$$

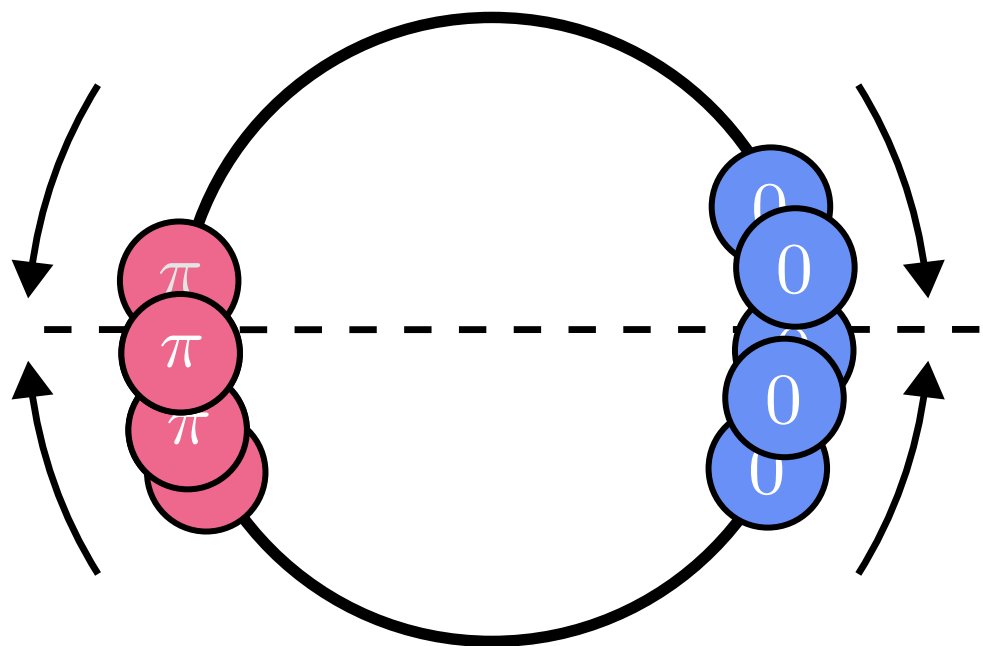
$$\partial_t \rho(\theta) = -\partial_\theta (\rho(\theta) v(\theta)) = -\partial_\theta \left(\rho(\theta) [K (\sin \star \rho)(\theta) - K_s \sin(2\theta)] \right)$$

- Gradient flow w/r/t 2-Wasserstein structure on space of phase densities

Eulerian Mean-Field Limit



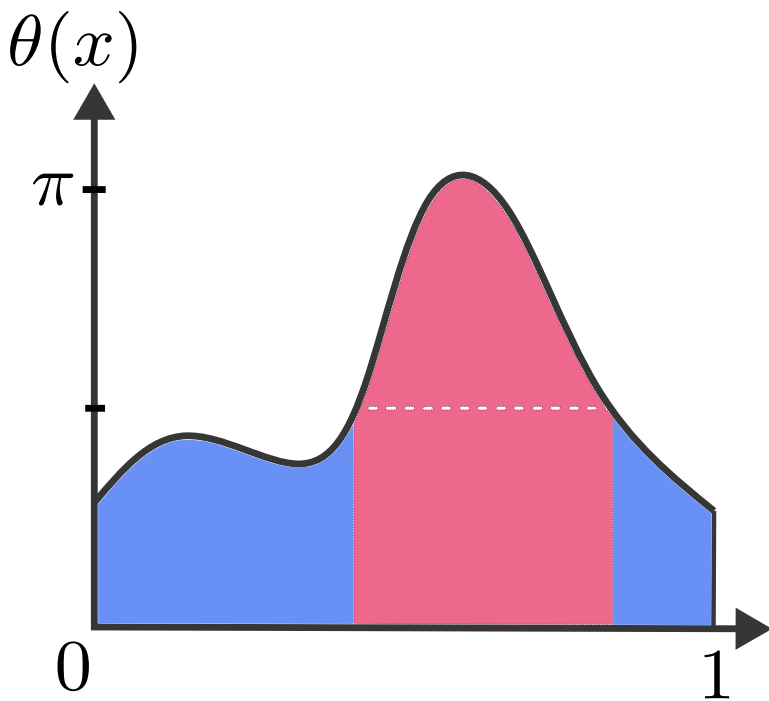
Oscillator Ising Machines



$$\dot{\theta}_i = \sum_j A_{ij} \sin(\theta_i - \theta_j) - K_s \sin(2\theta_i)$$

$$A = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix}$$

Lagrangian formulation

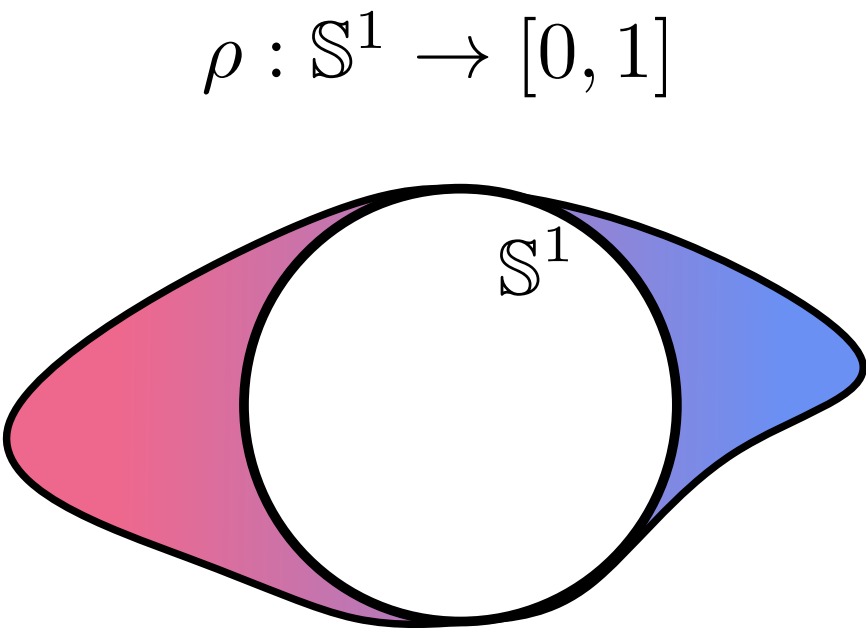


$$\partial_t \theta(x) = \int K(x, y) \sin(\theta(x) - \theta(y)) \, dy - K_s \sin(2\theta(x))$$



$$K : [0, 1]^2 \rightarrow \mathbb{R}$$

Eulerian formulation



$$\partial_t \rho(\theta) = -\nabla \cdot (\rho(\theta) [K(\sin * \rho)(\theta) - K_s \sin(2\theta)])$$



$$K : [0, 1]^2 \rightarrow \mathbb{R}$$

Main Problem

Problem: Classify fixed points and stability of mean-field models:

Lagrangian :
$$\partial_t \boldsymbol{\theta}(x) = K \int \sin(\boldsymbol{\theta}(x) - \boldsymbol{\theta}(y)) dy - K_s \sin(2\boldsymbol{\theta}(x))$$

Eulerian :
$$\partial_t \boldsymbol{\rho}(\theta) = - \partial_\theta \left(\boldsymbol{\rho}(\theta) \left[K (\sin \star \boldsymbol{\rho})(\theta) - K_s \sin(2\theta) \right] \right)$$

Approach: Use the gradient-flow structure, compute ∇E , $\nabla^2 E$, etc.

Equilibrium Conditions

Fixed point conditions:

$$\nabla E(\boldsymbol{\theta})(x) = -K \int \sin(\boldsymbol{\theta}(x) - \boldsymbol{\theta}(y)) dy + K_s \sin(2\boldsymbol{\theta}(x)) \equiv 0$$

$\Rightarrow \boldsymbol{\theta}$ valued in (ρ supported on) zeros of vector field

$$v_{\boldsymbol{\theta}}(\theta) = -K \int \sin(\theta - \boldsymbol{\theta}(y)) dy + K_s \sin(2\theta)$$

$\Rightarrow$ phases of oscillators can only cluster at zeros

Equilibrium Conditions

Vector field v admits finite parameterization via order parameter z :

$$v_z(\theta) = -K \operatorname{Im}(ze^{-i\theta}) - K_s \sin(2\theta),$$

$$z := \int_I e^{i\theta(x)} dx$$

$\Rightarrow$ can be shown that v has either 2, 3, or 4 zeros

$\Rightarrow$ oscillator phases cluster into either 1, 2, 3, or 4 groups

Stability Conditions

Stability conditions:

$$\begin{aligned} D^2E(\boldsymbol{\theta})[p, p] &= \int \left(2K_s \cos(2\boldsymbol{\theta}(x)) - K \int \cos(\boldsymbol{\theta}(x) - \boldsymbol{\theta}(y)) dy \right) p^2(x) dx \\ &\quad + K \iint \cos(\boldsymbol{\theta}(x) - \boldsymbol{\theta}(y)) p(y) p(x) dy dx \geq 0 \quad \forall p \end{aligned}$$

⇒ Can be shown fixed points with clusters outside of $\{0, \pi\}$ unstable

⇒ Feasible solutions to original problem almost always recoverable

Stability Conditions

⇒ Binarized fixed points stable when:

$$K/K_s \leq 2$$

$$m \in \{0,1\},$$

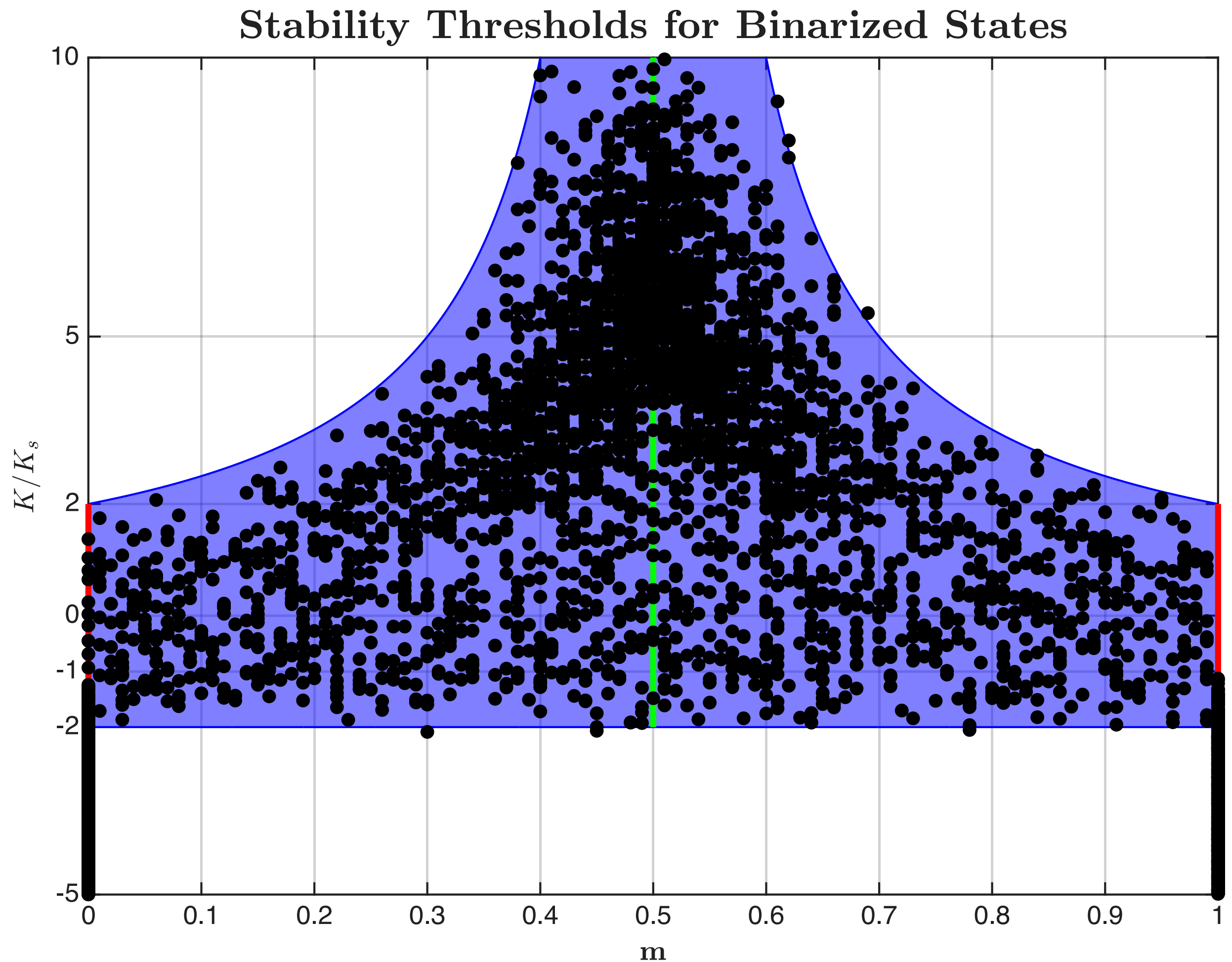
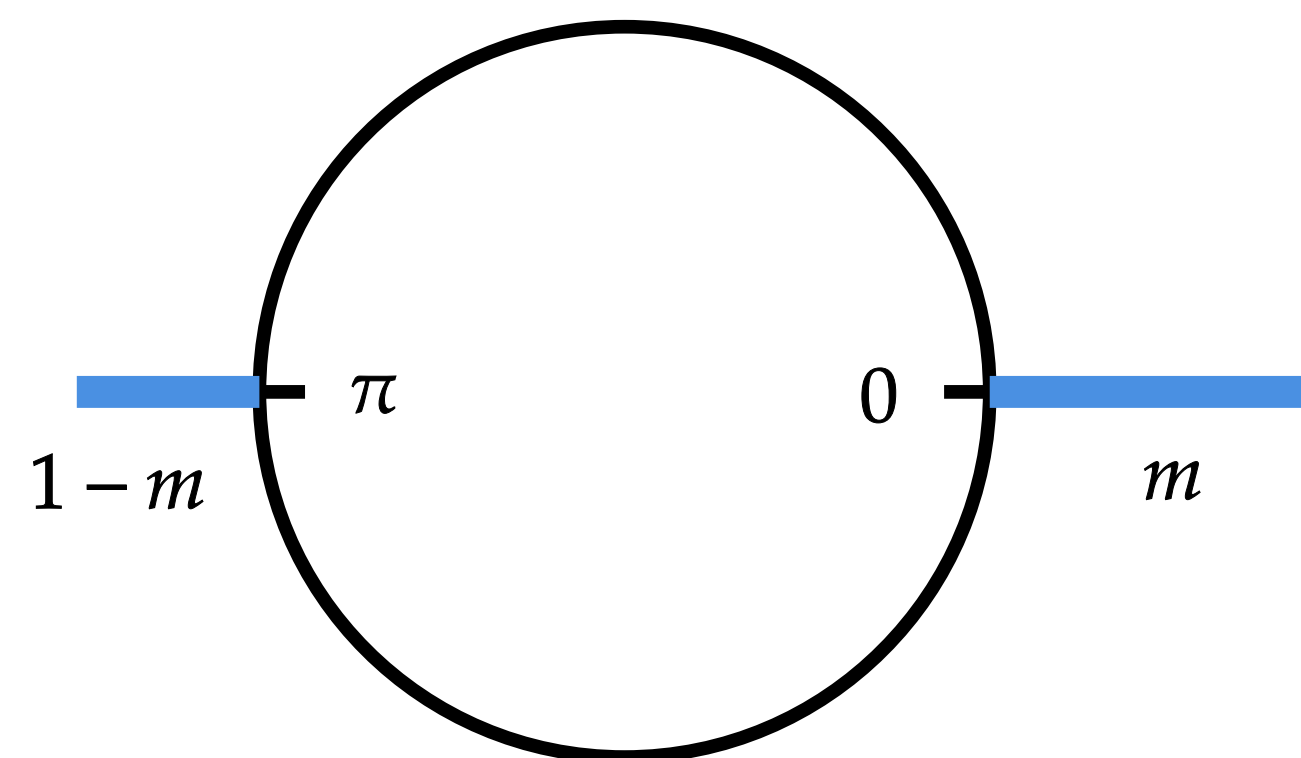
$$-2 \leq K/K_s$$

$$m = 0.5,$$

$$-2 \leq K/K_s \leq \frac{2}{|2m - 1|} \quad \text{otherwise,}$$

where $m \in [0,1]$ is mass fraction at 0

⇒ Threshold for binarization is $-2 \leq K/K_s$



Looking Forward

- Big Question: How far can the analysis be extended?
- More general OIM models:

$$\dot{\theta}_i = \sum_j \underbrace{A_{ij}}_{\text{structured}} \sin(\theta_i - \theta_j) - \underbrace{K_i(t)}_{\text{free}} \sin(2\theta_i) + \underbrace{w_i(t)}_{\text{noise}}, \quad \theta_i(0) \sim \underbrace{X_i}_{\text{free}}$$

- Do these models retain gradient flow structures? Are they tractable?
- What is the relationship between frequency and quality of fixed points?
- How to choose free parameters to get best performance?

Thanks for watching! Questions?



Link to Paper



Personal Website